

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I _M Kavya 192472370__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: kavya

CASE STUDY

CASE STUDY 1 — SMART MEDICAL DIAGNOSIS SYSTEM

(a) Representation using Propositional Logic

Let the following propositional symbols represent the facts:

- **F** = Patient A has Fever
- **C** = Patient A has Cough
- **B** = Patient A has Breathlessness
- **Flu** = Patient A has Possible Flu
- **M** = Patient A has Possible Measles
- **P** = Patient A has Possible Pneumonia
- **R** = Patient A has Rash

The given rules can be represented as follows:

Rule I:

If a patient has Fever AND Cough, then Possible Flu.

$F \wedge C \rightarrow \text{Flu}$

Rule II:

If a patient has Fever AND Rash, then Possible Measles.

$F \wedge R \rightarrow M$

Rule III:

If a patient has Cough AND Breathlessness, then Possible Pneumonia.

$C \wedge B \rightarrow P$

Facts about Patient A:

Therefore, the initial knowledge base is:

$F = \text{True}$

$C = \text{True}$

$B = \text{True}$

$KB = \{ F, C, B, (F \wedge C \rightarrow \text{Flu}), (F \wedge R \rightarrow M), (C \wedge B \rightarrow P) \}$

(b) Forward Chaining

Forward chaining starts with the known facts and repeatedly applies rules whose conditions are satisfied.

Initial Facts:

Fever(A)=true

Cough(A)=True

Breathlessness(A)=True

Step 1 – Apply Rule I

Rule I:

$F \wedge C \rightarrow \text{Flu}$

Since both Fever and Cough are true:

FEVER^COUGH=True

Therefore:

Possible Flu(A) = True

Rule Fired: Rule I

New Fact: Possible Flu

Step 2 – Apply Rule II

Rule II:

Fever ^ Rash - Measles

We know Fever is true, but there is **no fact stating that Patient A has Rash**.

Therefore, Rule II cannot be fired.

Conclusion: Possible Measles cannot be derived.

Step 3 – Apply Rule III

Rule III:

Cough ^ Breathlessness - pneumonia

Both Cough and Breathlessness are true.

Therefore:

Possible Pneumonia (A) = True

Rule Fired: Rule III

New Fact: Possible Pneumonia

Forward Chaining Summary

Step	Known Facts	Rule Fired	New Fact
1	Fever, Cough	Rule I	Possible Flu
2	Fever, no Rash	Rule II not fired	No Measles
3	Cough, Breathlessness	Rule III	Possible Pneumonia

Final Conclusions

Possible Measles cannot be derived because Rash is not known.

(c) Backward Chaining for Pneumonia

Goal:

Pneumonia(A)

Backward chaining starts from the goal and searches for rules that can derive it.

The only rule whose conclusion is Pneumonia is:

Cough ^ Breathlessness - Pneumonia

Therefore, the goal is reduced into two sub-goals:

Pneumonia(A)

↓

Cough(A) ^ Breathlessness(A)

Both sub-goals are present as facts:

Cough(A)=True

Breathlessness(A)=True

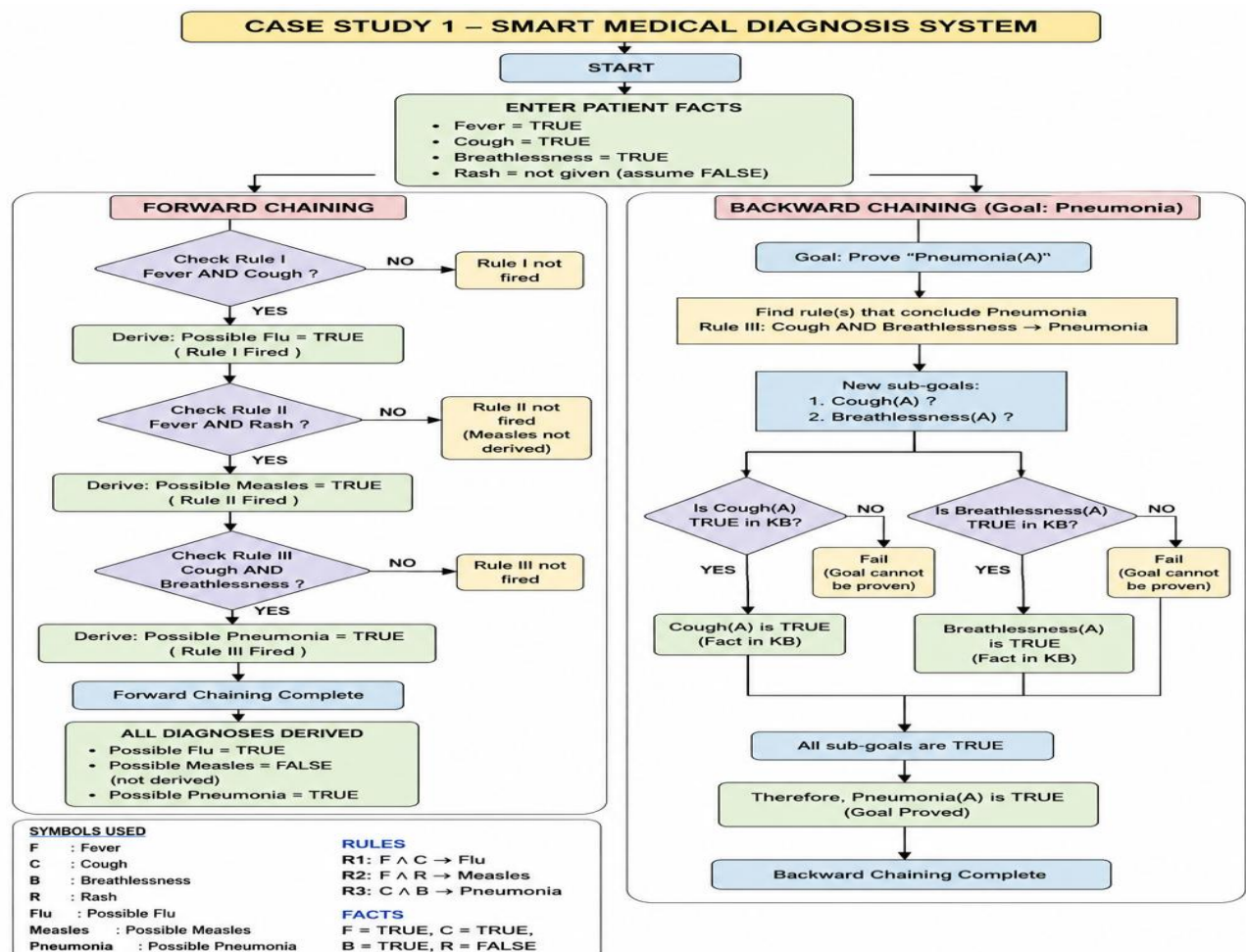
Therefore:

$Cough(A) \wedge Breathlessness(A) = True$

Hence:

Pneumonia (A) = True

Goal Reduction Tree



Conclusion

Backward chaining successfully verifies that **Patient A has Possible Pneumonia**, because both required conditions, Cough and Breathlessness, are available as facts.

CASE STUDY 2 — AUTONOMOUS TRAFFIC MANAGEMENT AGENT

(a) Apply Unification to Match Rule 1 with the Given Facts

The given Rule 1 is:

$\forall x: Vehicle(x) \wedge EmergencyType(x) \rightarrow ClearPath(x)$

The given facts are:

Vehicle(Ambulance)

EmergencyType(Ambulance)

The rule contains the variable **x**. We need to find a substitution that makes the rule match the given facts.

Step 1: Match Vehicle(x)

Match:

Vehicle(x)

with:

Vehicle(Ambulance)

Therefore, the substitution is:

$\theta_1 = \{x/\text{Ambulance}\}$

Step 2: Match EmergencyType(x)

Substitute $x = \text{Ambulance}$ into the second condition:

EmergencyType(Ambulance)

This fact is also available in the knowledge base.

Therefore:

$\theta_2 = \{x/\text{Ambulance}\}$

Most General Unifier (MGU)

The complete substitution is:

$\theta = \{x/\text{Ambulance}\}$

After applying the substitution to Rule 1:

$\text{Vehicle(Ambulance)} \wedge \text{EmergencyType(Ambulance)} \rightarrow \text{ClearPath(Ambulance)}$

Since both conditions are true:

ClearPath(Ambulance)

Therefore:

ClearPath(Ambulance) is derived.

(b) Using Resolution Method, Prove that Proceed(CarA) is True

The goal is:

Proceed(CarA)

We convert the relevant rules into Conjunctive Normal Form (CNF).

Rule 1

Original:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Eliminating implication:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Therefore:

C1: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2

Original:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

This gives two CNF clauses:

C2: $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

C3: $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

For proving $\text{Proceed}(\text{CarA})$, only C2 is required.

Rule 3

Original:

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

CNF form:

C4: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Given Facts

C5: $\text{Vehicle}(\text{Ambulance})$

C6: $\text{EmergencyType}(\text{Ambulance})$

C7: $\text{Vehicle}(\text{CarA})$

C8: $\text{Behind}(\text{CarA}, \text{Ambulance})$

Resolution Step 1

Resolve C1 with C5.

C1:

$\neg \text{Vehicle}(\text{Ambulance}) \vee \neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$

C5:

$\text{Vehicle}(\text{Ambulance})$

After resolution:

R1:

$\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$

Resolution Step 2

Resolve R1 with C6:

R1:

$\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$

C6:

$\text{EmergencyType}(\text{Ambulance})$

Therefore:

R2:

$\text{ClearPath}(\text{Ambulance})$

Resolution Step 3

Resolve C2 with R2.

C2:

$\neg \text{ClearPath}(\text{Ambulance}) \vee \text{GreenSignal}(\text{Ambulance})$

R2:

$\text{ClearPath}(\text{Ambulance})$

Therefore:

R3:

GreenSignal(Ambulance)

Resolution Step 4

Use Rule 3 with:

$x = \text{CarA}$

$y = \text{Ambulance}$

C4 becomes:

$\neg \text{Vehicle}(\text{CarA}) \vee \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

Resolve with C7:

$\text{Vehicle}(\text{CarA})$

Therefore:

R4:

$\neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

Resolution Step 5

Resolve R4 with C8:

C8:

$\text{Behind}(\text{CarA}, \text{Ambulance})$

Therefore:

R5:

$\neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

Resolution Step 6

Resolve R5 with R3:

R5:

$\neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

R3:

$\text{GreenSignal}(\text{Ambulance})$

Therefore:

R6:

$\text{Proceed}(\text{CarA})$

Hence:

$\text{Proceed}(\text{CarA})$ is logically derived.

Therefore:

$\text{KB} \models \text{Proceed}(\text{CarA})$

Resolution Proof Tree

CASE STUDY 2 – Resolution Chain to Prove PROCEED(CAR A)

1. KNOWLEDGE BASE CONVERTED TO CNF

RULES CONVERTED TO CLAUSES	FACTS (as unit clauses)
C1: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$	C5: $\text{Vehicle}(\text{Ambulance})$
C2: $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$	C6: $\text{EmergencyType}(\text{Ambulance})$
C3: $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$	C7: $\text{Vehicle}(\text{CarA})$
C4: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$	C8: $\text{Behind}(\text{CarA}, \text{Ambulance})$

2. RESOLUTION CHAIN

STEP	CLAUSES RESOLVED	LITERAL RESOLVED	RESOLVENT (NEW CLAUSE)	JUSTIFICATION
①	C1: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$ C5: $\text{Vehicle}(\text{Ambulance})$	$\neg \text{Vehicle}(x)$ $\text{Vehicle}(\text{Ambulance})$	R1: $\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$	x / Ambulance (Resolve on Vehicle)
②	R1: $\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$ C6: $\text{EmergencyType}(\text{Ambulance})$	$\neg \text{EmergencyType}(\text{Ambulance})$ $\text{EmergencyType}(\text{Ambulance})$	R2: $\text{ClearPath}(\text{Ambulance})$	(Resolve on EmergencyType)
③	C2: $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$ R2: $\text{ClearPath}(\text{Ambulance})$	$\neg \text{ClearPath}(x)$ $\text{ClearPath}(\text{Ambulance})$	R3: $\text{GreenSignal}(\text{Ambulance})$	x / Ambulance (Resolve on ClearPath)
④	C4: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$ C7: $\text{Vehicle}(\text{CarA})$	$\neg \text{Vehicle}(x)$ $\text{Vehicle}(\text{CarA})$	R4: $\neg \text{Behind}(\text{CarA},y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(\text{CarA})$	x / CarA (Resolve on Vehicle)
⑤	R4: $\neg \text{Behind}(\text{CarA},y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(\text{CarA})$ C8: $\text{Behind}(\text{CarA}, \text{Ambulance})$	$\neg \text{Behind}(\text{CarA},y)$ $\text{Behind}(\text{CarA}, \text{Ambulance})$	R5: $\neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$	y / Ambulance (Resolve on Behind)
⑥	R5: $\neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$ R3: $\text{GreenSignal}(\text{Ambulance})$	$\neg \text{GreenSignal}(\text{Ambulance})$ $\text{GreenSignal}(\text{Ambulance})$	R6: $\text{Proceed}(\text{CarA})$	(Resolve on GreenSignal)

FINAL RESULT

R6: $\text{Proceed}(\text{CarA})$ is derived.

Therefore, $\text{KB} \models \text{Proceed}(\text{CarA})$ (Proved by Resolution)

(c) Analyze Whether the Current FOL Knowledge Base is Sufficient for Two Simultaneous Emergency Vehicles

The current FOL knowledge base can reason about more than one emergency vehicle if each vehicle has the required facts.

For example, suppose there are two emergency vehicles:

$\text{Vehicle}(\text{Ambulance1})$

$\text{EmergencyType}(\text{Ambulance1})$

$\text{Vehicle}(\text{Ambulance2})$

$\text{EmergencyType}(\text{Ambulance2})$

Using Rule 1:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

the system can derive:

ClearPath(Ambulance1)

and:

ClearPath(Ambulance2)

Using Rule 2:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

the system can derive:

GreenSignal(Ambulance1)

and:

GreenSignal(Ambulance2)

Therefore, the existing rules can independently process multiple emergency vehicles.

However, the current knowledge base is **not completely sufficient when the two emergency vehicles compete for the same road, lane, or traffic signal**. There is no rule specifying which emergency vehicle should receive priority.

Additional Knowledge Required

The system should include facts and rules related to:

1. Emergency vehicle priority
2. Emergency severity
3. Road or intersection conflicts
4. Signal allocation
5. Vehicle ordering
6. Collision avoidance

For example, an additional rule can be:

$\text{EmergencyVehicle}(x) \wedge \text{EmergencyVehicle}(y) \wedge \text{Conflict}(x,y) \wedge \text{Priority}(x,y) \rightarrow \text{GivePriority}(x)$

A fact can specify:

Priority(Ambulance1,Ambulance2)

This means Ambulance1 has priority over Ambulance2.

Another useful rule could be:

$\text{GivePriority}(x) \wedge \text{Intersection}(i) \rightarrow \text{GreenSignalFor}(x,i)$

This allows the system to assign the traffic signal to the higher-priority emergency vehicle.

Logical Analysis

The current knowledge base is sufficient to determine that an individual emergency vehicle can receive a clear path and green signal. It is also capable of applying the rules independently to multiple vehicles.

However, it does not contain sufficient knowledge for resolving conflicts between simultaneous emergency vehicles.

Therefore:

The current KB is partially sufficient for two simultaneous emergency vehicles, but additional priority and conflict-resolution rules are required for complete traffic management.

Final Conclusion

The FOL knowledge base successfully demonstrates:

- Unification of variables with constants.
- Derivation of ClearPath(Ambulance).
- Derivation of GreenSignal(Ambulance).
- Resolution-based proof of Proceed(CarA).
- Logical reasoning about multiple emergency vehicles.

However, for a real-world autonomous traffic management system, additional rules for **priority, conflict detection, road allocation, and emergency severity** are required.

CASE STUDY 3 — AGRICULTURAL EXPERT REASONING SYSTEM

(a) Convert P1, P2, P3 and Facts into Conjunctive Normal Form (CNF)

The given propositional knowledge base contains the following rules:

P1:

SoilDry $\rightarrow$ IrrigationNeeded

P2:

IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

P3:

$\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Facts:

SoilDry = True

CropWheat = True

Conversion of P1 into CNF

Original statement:

SoilDry $\rightarrow$ IrrigationNeeded

Using the implication equivalence:

$A \rightarrow B \equiv \neg A \vee B$

Therefore:

$\neg$ SoilDry $\vee$ IrrigationNeeded

Hence, the CNF form is:

C1: $\neg$ SoilDry $\vee$ IrrigationNeeded

Conversion of P2 into CNF

Original statement:

IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Eliminate the implication:

$\neg$ (IrrigationNeeded $\wedge$ CropWheat) $\vee$ ApplyDripMethod

Using De Morgan's Law:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Hence, the CNF form is:

C2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Conversion of P3 into CNF

Original statement:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Eliminate the implication:

$\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Move the negation inward using De Morgan's Law:

$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Hence, the CNF form is:

C3: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Conversion of Facts into CNF

Given:

$\text{SoilDry} = \text{True}$

Therefore:

C4: SoilDry

Given:

$\text{CropWheat} = \text{True}$

Therefore:

C5: CropWheat

Complete CNF Knowledge Base

The complete CNF representation is:

C1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

C2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

C3: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

C4: SoilDry

C5: CropWheat

These clauses form the complete CNF knowledge base.

(b) Resolution Refutation to Prove ApplyDripMethod

The theorem to be proved is:

ApplyDripMethod

In Resolution Refutation, the negation of the goal is added to the knowledge base.

Therefore, the negated goal is:

C6: $\neg \text{ApplyDripMethod}$

We now combine C6 with the CNF knowledge base and derive the empty clause $\square$.

Resolution Step 1: Derive IrrigationNeeded

C1:

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

C4:

SoilDry

Resolving on SoilDry:

R1: IrrigationNeeded

Resolution Step 2: Derive ApplyDripMethod

C2:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

R1:

IrrigationNeeded

Resolving on IrrigationNeeded:

R2: $\neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Resolution Step 3: Use CropWheat

R2:

$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$

C5:

CropWheat

Resolving on CropWheat:

R3: ApplyDripMethod

Resolution Step 4: Derive the Empty Clause

We added the negated goal:

C6:

$\neg \text{ApplyDripMethod}$

We have already derived:

R3:

ApplyDripMethod

Resolve R3 and C6:

Therefore, the negated goal $\neg \text{ApplyDripMethod}$ is false.

Hence, the original theorem is proven:

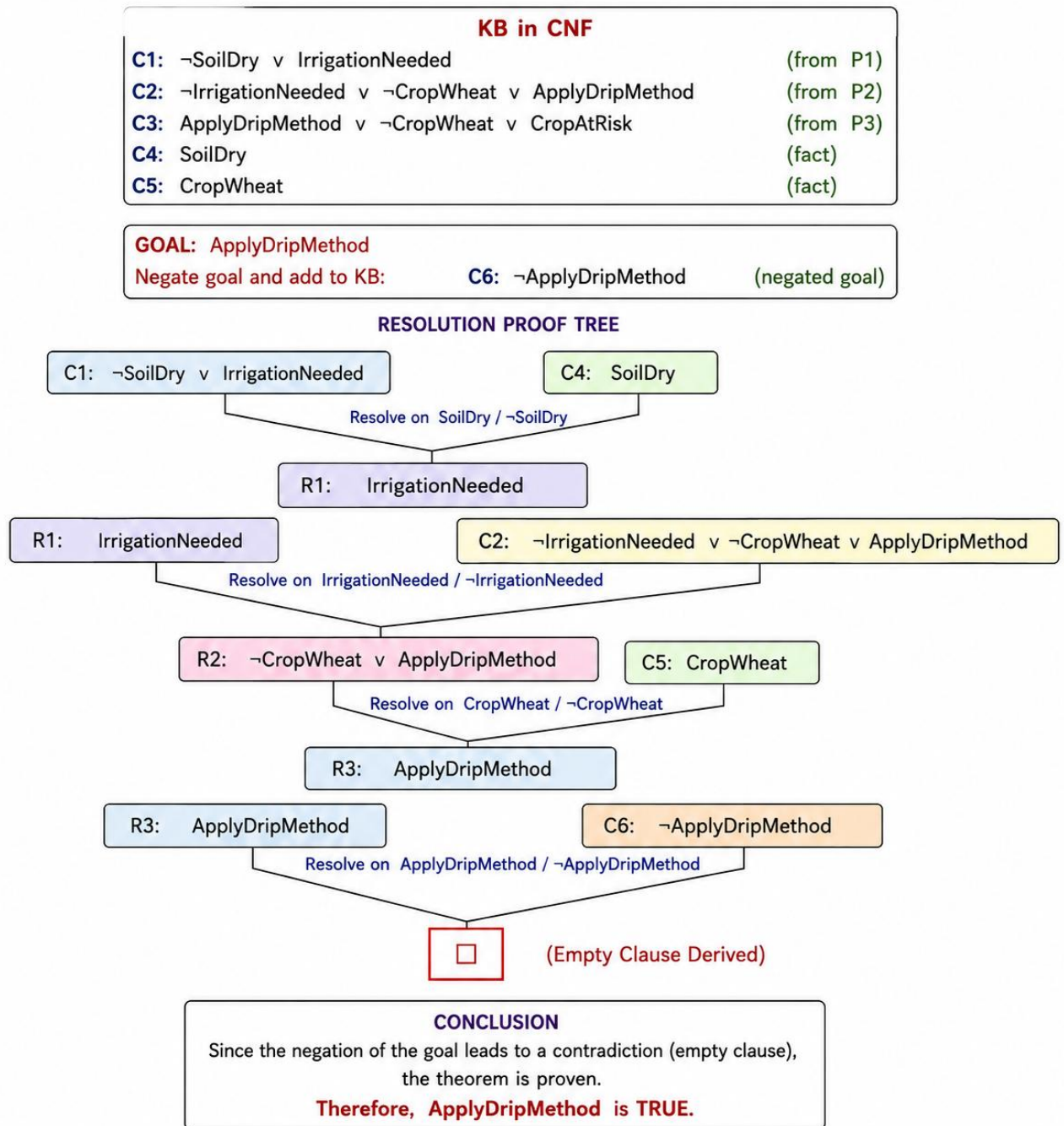
KB $\models$ ApplyDripMethod

Therefore:

ApplyDripMethod is TRUE.

Resolution Proof Tree

CASE STUDY 3 – RESOLUTION PROOF TREE TO PROVE ApplyDripMethod



Conclusion

Since the negation of the goal leads to the empty clause:

$\text{KB} \cup \{\neg \text{ApplyDripMethod}\} \vdash \square$

Therefore:

$\text{KB} \models \text{ApplyDripMethod}$

Thus, the agricultural reasoning system logically concludes that:

ApplyDripMethod = TRUE

(c) Effect of Removing SoilDry from the Knowledge Base

Suppose the fact:

SoilDry = True

is removed from the knowledge base.

The remaining facts are:

CropWheat = True

The rules remain:

P1:

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

P2:

$\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

P3:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Step 1: Check P1

P1 states:

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Since SoilDry is no longer available as a fact, the system cannot derive:

IrrigationNeeded

Therefore:

IrrigationNeeded cannot be concluded.

Step 2: Check P2

P2 states:

$\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Although CropWheat is still true, IrrigationNeeded is not known.

Therefore:

ApplyDripMethod cannot be derived.

Step 3: Check P3

P3 states:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

We know:

CropWheat = True

However, the knowledge base does not contain:

$\neg \text{ApplyDripMethod}$

The absence of ApplyDripMethod does not mean that $\neg \text{ApplyDripMethod}$ is true.

Therefore:

CropAtRisk cannot be logically concluded.

Final Analysis

Fact/Conclusion	Before removing SoilDry	After removing SoilDry
SoilDry	True	Unknown
IrrigationNeeded	True	Cannot derive
CropWheat	True	True
ApplyDripMethod	True	Cannot derive
$\neg$ ApplyDripMethod	Not given	Still not given
CropAtRisk	Cannot derive	Cannot derive

Conclusion

After removing SoilDry, the system can no longer derive IrrigationNeeded or ApplyDripMethod. However, **CropAtRisk also does not follow**, because there is no explicit fact establishing .

CASE STUDY 4 — WUMPUS WORLD KNOWLEDGE AGENT

(a) Construct the Agent's Belief State and Apply Modus Ponens

The agent perceives the following:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

The given rules are:

Rule 5:

Stench[1,2] $\rightarrow$ WumpusAdjacent[1,2]

Rule 6:

Breeze[1,1] $\rightarrow$ PitAdjacent[1,1]

Rule 7:

Glitter[x,y] $\rightarrow$ Gold[x,y]

Rule 8:

$\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

Step 1: Apply Rule 5

Given:

Stench[1,2]

Rule 5:

Stench[1,2] $\rightarrow$ WumpusAdjacent[1,2]

Using Modus Ponens:

Stench[1,2]

-

$\text{Stench}[1,2] \rightarrow \text{WumpusAdjacent}[1,2]$

Therefore:

WumpusAdjacent[1,2]

This means a Wumpus may be located in a cell adjacent to [1,2].

Step 2: Apply Rule 6

Given:

Breeze[1,1]

Rule 6:

$\text{Breeze}[1,1] \rightarrow \text{PitAdjacent}[1,1]$

Using Modus Ponens:

Breeze[1,1]

•

$\text{Breeze}[1,1] \rightarrow \text{PitAdjacent}[1,1]$

Therefore:

PitAdjacent[1,1]

This indicates that a pit may be present in a cell adjacent to [1,1].

Step 3: Apply Rule 7

Given:

Glitter[2,2]

Rule 7:

$\text{Glitter}[x,y] \rightarrow \text{Gold}[x,y]$

Substituting:

$x = 2, y = 2$

Therefore:

Gold[2,2]

The agent has confirmed that the gold is at cell [2,2].

Step 4: Check Rule 8

Rule 8 states:

$\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

To apply this rule, the agent must know both:

$\neg \text{Wumpus}[x,y]$

and:

$\neg \text{Pit}[x,y]$

No such negative facts are provided.

Therefore, Rule 8 cannot be fired.

Belief State After Applying Modus Ponens

The agent's derived knowledge is:

1. Stench[1,2]
2. Breeze[1,1]
3. Glitter[2,2]
4. WumpusAdjacent[1,2]
5. PitAdjacent[1,1]
6. Gold[2,2]

Therefore, the agent knows the possible presence of a Wumpus near [1,2], a possible pit near [1,1], and the location of the gold at [2,2].

(b) Forward Chaining to Identify Possible Locations of Wumpus and Pit

Forward Chaining starts with the known percepts and applies the rules to derive new knowledge.

Step 1: Identify Possible Wumpus Locations

Initial fact:

Stench[1,2]

Apply Rule 5:

Stench[1,2] $\rightarrow$ WumpusAdjacent[1,2]

Therefore:

WumpusAdjacent[1,2]

In a standard Wumpus World grid, the cells adjacent to [1,2] can include:

[1,1], [1,3], [2,2]

Therefore, the possible Wumpus locations are:

[1,1], [1,3], and [2,2]

The exact Wumpus location cannot be determined using the given information alone.

Step 2: Identify Possible Pit Location

Initial fact:

Breeze[1,1]

Apply Rule 6:

Breeze[1,1] $\rightarrow$ PitAdjacent[1,1]

Therefore:

PitAdjacent[1,1]

Since [1,1] is a corner cell in the standard grid, its adjacent cell is:

[1,2]

Therefore:

[1,2] is a possible Pit location.

Step 3: Identify Gold Location

Initial fact:

Glitter[2,2]

Apply Rule 7:

Glitter[2,2] $\rightarrow$ Gold[2,2]

Therefore:

Gold is located at [2,2].

Forward Chaining Summary

Initial Percept	Rule Applied	Derived Knowledge
Stench [1,2]	Rule 5	Wumpus adjacent to [1,2]
Breeze [1,1]	Rule 6	Pit adjacent to [1,1]
Glitter [2,2]	Rule 7	Gold at [2,2]

Final Possible Locations

Possible Wumpus locations:

[1,1], [1,3], [2,2]

Possible Pit location:

[1,2]

Gold location:

[2,2]

(c) Evaluate Whether the Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is Safe

The proposed path is:

[1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

The agent must determine whether each cell can be logically considered safe.

Cell [1,1]

The agent perceives:

Breeze[1,1]

Using Rule 6:

Breeze[1,1] $\rightarrow$ PitAdjacent[1,1]

This means a pit may exist in an adjacent cell.

Therefore, the agent does not have enough information to prove that all neighboring cells are safe.

Cell [1,2]

The agent knows:

Stench[1,2]

Using Rule 5:

Stench[1,2] $\rightarrow$ WumpusAdjacent[1,2]

Also, from the Breeze at [1,1], [1,2] is a possible pit location.

Therefore, [1,2] cannot be confirmed as safe.

Cell [2,2]

The agent perceives:

Glitter[2,2]

Therefore:

Gold[2,2]

The gold is confirmed to be at [2,2].

However, glitter does not prove that the cell is safe.

According to Rule 8:

$\neg \text{Wumpus}[2,2] \wedge \neg \text{Pit}[2,2] \rightarrow \text{Safe}[2,2]$

The knowledge base does not provide:

$\neg \text{Wumpus}[2,2]$

and:

$\neg \text{Pit}[2,2]$

Therefore:

Safe[2,2]

cannot be derived.

Path Safety Analysis

Cell	Available Knowledge	Safety
[1,1]	Breeze detected	Not fully confirmed safe
[1,2]	Possible Pit; Wumpus-related Stench	Not safe to confirm
[2,2]	Gold confirmed	Safety not established

Therefore, the complete path:

[1,1] → [1,2] → [2,2]

cannot be proven safe using the current knowledge base.

Final Conclusion

Although the agent can confirm that:

Gold[2,2]

it cannot confirm:

Safe[1,2]

or:

Safe[2,2]

because the required negative facts about the Wumpus and Pit are not available.

Therefore:

The proposed path cannot be guaranteed safe with the available knowledge.

The rational agent should obtain additional percepts or knowledge before traversing the complete path.

CASE STUDY 4 – WUMPUS WORLD

FORWARD CHAINING SUMMARY

INITIAL PERCEPTS (FACTS)		RULES IN THE KNOWLEDGE BASE		
- Stench at [1,2] - Breeze at [1,1] - Glitter at [2,2]		R5: Stench [1,2] → Wumpus is adjacent to cell [1,2] R6: Breeze [1,1] → Pit is adjacent to cell [1,1] R7: Glitter [x,y] → Gold is at [x,y] R8: ¬Wumpus [x,y] ∧ ¬Pit [x,y] → Safe [x,y]		
STEP-BY-STEP FORWARD CHAINING				
Step	Known Facts	Rule Applied	New Fact Derived	Explanation
0	- Stench [1,2] - Breeze [1,1] - Glitter [2,2]	–	–	Initial percepts from the agent's sensors.
1	- Stench [1,2] - Breeze [1,1] - Glitter [2,2]	R5: Stench [1,2] → Wumpus adjacent to [1,2]	WumpusAdjacent [1,2]	From Stench at [1,2], we infer Wumpus is in one of the cells adjacent to [1,2]. Possible locations: [1,1] , [1,3] , [2,2]
2	- Stench [1,2] - Breeze [1,1] - Glitter [2,2] - WumpusAdjacent [1,2]	R6: Breeze [1,1] → Pit adjacent to [1,1]	PitAdjacent [1,1]	From Breeze at [1,1], we infer Pit is in one of the cells adjacent to [1,1]. (For standard grid: [1,2])
3	- Stench [1,2] - Breeze [1,1] - Glitter [2,2] - WumpusAdjacent [1,2] - PitAdjacent [1,1]	R7: Glitter [x,y] → Gold is at [x,y]	Gold [2,2]	From Glitter at [2,2], we infer Gold is at cell [2,2].
4	- Stench [1,2] - Breeze [1,1] - Glitter [2,2] - WumpusAdjacent [1,2] - PitAdjacent [1,1] - Gold [2,2]	R8: ¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]	No new fact derived	We do not have any cell with both ¬Wumpus and ¬Pit proven. Therefore, Safe[x,y] cannot be derived yet.

FORWARD CHAINING FLOW DIAGRAM		FINAL DERIVED KNOWLEDGE	
<pre>graph TD START([START]) --> PERCEPTS[Percepts: Stench[1,2], Breeze[1,1], Glitter[2,2]] PERCEPTS --> R5[Apply R5 Stench[1,2] → WumpusAdjacent[1,2]] PERCEPTS --> R6[Apply R6 Breeze[1,1] → PitAdjacent[1,1]] PERCEPTS --> R7[Apply R7 Glitter[2,2] → Gold[2,2]] R5 --> WUMPUS_POSS[Possible Wumpus Locations: [1,1] , [1,3] , [2,2]] R6 --> PIT_POSS[Possible Pit Location: [1,2]] R7 --> GOLD[Gold is at [2,2]] WUMPUS_POSS --> R8[Apply R8 ¬Wumpus ∧ ¬Pit → Safe] PIT_POSS --> R8 GOLD --> R8 R8 --> NO_SAFE[No Safe cell confirmed yet] NO_SAFE --> END([END])</pre>		- WumpusAdjacent [1,2] → Wumpus may be in: [1,1] , [1,3] , [2,2] - PitAdjacent [1,1] → Pit may be in: [1,2] (for standard grid) - Gold is at [2,2] - Safe cells: Cannot be confirmed (need ¬Wumpus and ¬Pit for a cell)	
		CONCLUSION	
		Using Forward Chaining, the agent can identify possible locations of the Wumpus and Pit and the exact location of Gold, but it cannot conclude any cell is safe without additional negative information.	

OVERALL CONCLUSION

The four case studies demonstrate different forms of logical reasoning used by Artificial Intelligence agents:

1. **Smart Medical Diagnosis** uses propositional logic, Forward Chaining, and Backward Chaining to derive possible diagnoses.
2. **Autonomous Traffic Management** uses FOL, unification, and resolution to reason about emergency vehicles and traffic signals.
3. **Agricultural Expert Reasoning** demonstrates CNF conversion and Resolution Refutation for proving logical conclusions.
4. **Wumpus World** demonstrates Modus Ponens and Forward Chaining for constructing an agent's belief state and evaluating safe actions.

These reasoning techniques allow intelligent agents to derive valid conclusions from known facts and rules, while also identifying situations where the available knowledge is insufficient for a definite conclusion.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand